

VITAL Workshop Tutorials

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March 2025

1 Software Setup

For setting up software, please follow the instructions detailed here: https://finbarargus.github.io/circulatory_autogen/getting-started/

2 0D Bond Graph Blood Flow Tutorial

Follow the demonstration on the whiteboard. Then complete the following tasks:

1. Write out the bond-graph diagram for a pressure inlet, pressure outlet vessel.
2. Derive the system of equations for the vessel. Knowns:
 - $C = 1 \cdot 10^{-8} \text{ m}^3 \cdot \text{Pa}^{-1}$
 - $R_1 = 1 \cdot 10^5 \text{ Pa} \cdot \text{s} \cdot \text{m}^{-3}$
 - $R_2 = 2 \cdot 10^5 \text{ Pa} \cdot \text{s} \cdot \text{m}^{-3}$
 - $I_1 = 10 \text{ Pa} \cdot \text{s}^2 \cdot \text{m}^{-3}$
 - $I_2 = 10 \text{ Pa} \cdot \text{s}^2 \cdot \text{m}^{-3}$
 - $P_{in} = [12 + 2 \sin(2\pi t)] \text{ kPa}$
 - $P_{out} = 1 \text{ kPa}$
3. Implement the equations in OpenCOR.
4. Plot the pressure at the centre of the vessel and the flow at the inlet.
5. Extra: Swap the tube law with the following:

$$P = P_0 + K \left(\left(\frac{q}{q_0} \right)^m - \left(\frac{q}{q_0} \right)^n \right) + P_{ext} \quad (1)$$

- Discuss whether (P_0, q_0) should be calibrated from data, or $P_0 = 0$ and $q_0 = q_{us}$ should be used. Simulate the differences in OpenCOR.

3 Circulatory System Generation Tutorial

1. Navigate to the 20250318-day01/simple_physiological_nonlinear_visco dir. We will call this dir {CA_user_dir} and the circulatory Autogen directory {project_dir}.
 - The CA_user_dir/resources directory contains input files that define the modules that are used in your models and their parameters (and later the information for parameter identification).
 - The project_dir contains src files for Circulatory Autogen and configuration files for modules (defining the mathematics and the allowable connections for the modules).

2. Open {project_dir}/user_run_files/user_inputs.yaml and set

```
user_inputs_path_override: {CA_user_dir}/simple_physiological_user_inputs.yaml
```

Important: For Windows, make sure you use “C:” at the start of the directory and backslash instead of forward-slash.

This lets CA know where to look for config files for the model you want to generate and eventually calibrate. The files that are needed to generate this circulatory system model are:

- `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_vessel_array.csv`: This file defines the modules that are used and how they are connected together. Figure 1 shows the model that we will first generate.

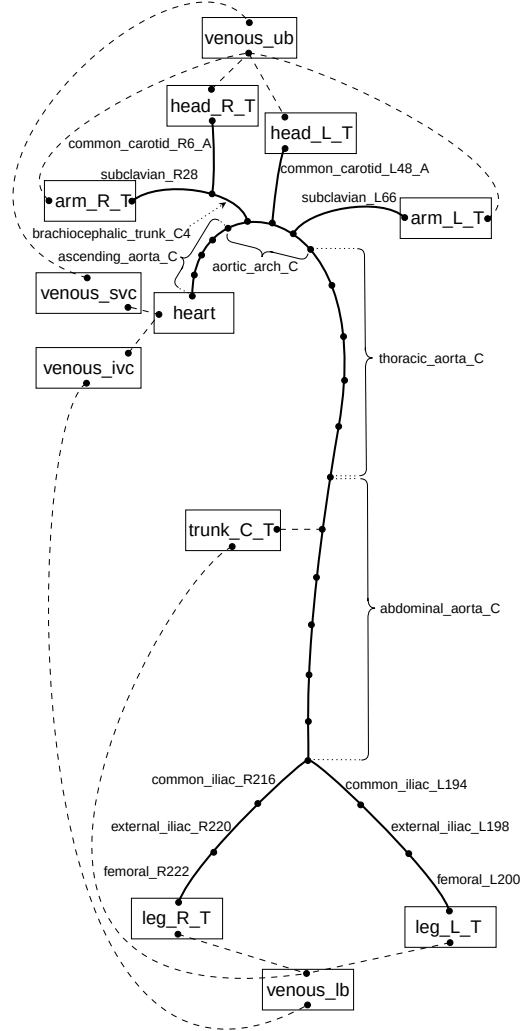


Figure 1: Schematic of the simple physiological CVS model.

- `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_parameters.csv`: This file defines the constants and boundary conditions needed to simulate the model.
- `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_params_for_id.csv`: This file defines the parameters that will be identified in the calibration steps (not modified in this tutorial but explained in https://finbarargus.github.io/circulatory_autogen/parameter-identification/).
- `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_obs_data.json`: This file defines the experimental/clinical protocols and the ground truth values that you want your outputs to replicate (also not modified in this tutorial).
- `{Ca_user_dir}/simple_physiological_user_inputs.yaml`: This is the main configuration file for your model, it specifies the location of your input files, the solver info, the parameter identification algorithm info, and more (you shouldn't need to edit it for this tutorial).

3. Navigate to `{project_dir}/user_run_files/` and execute the following command to generate the model:

```
./run_autogeneration.sh
```

The model will be generated in

```
{CA_user_dir}/generated_models/simple_physiological_nonlinear_visco
```

4. Open the generated model in OpenCOR (`simple_physiological_nonlinear_visco.cellml`) and plot the left subclavian pressure: `subclavian_L66/u`.
5. Now, assuming that we want to use PPG measurements from the finger to calibrate the model, we want to extend the vasculature network so that it includes vessels down to the fingers. The network we will include (for the left arm only) is displayed in Figure 2.

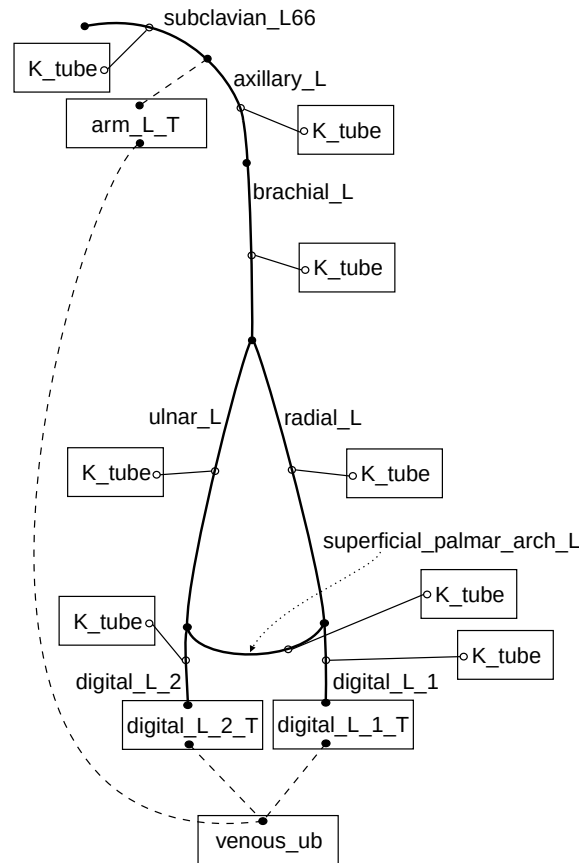


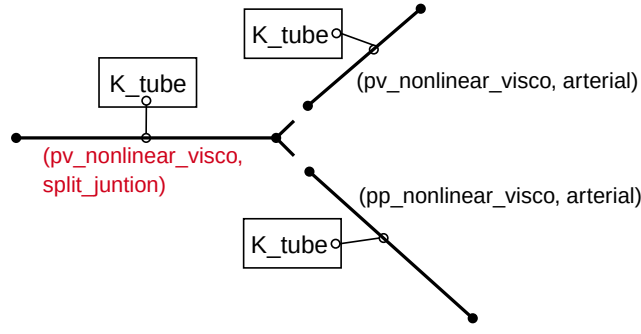
Figure 2: Schematic of the network that you need to include in the circulatory system model.

6. Modify the `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_vessel_array.csv` and `{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_parameters.csv` files so that they include the vessels in Figure 2.

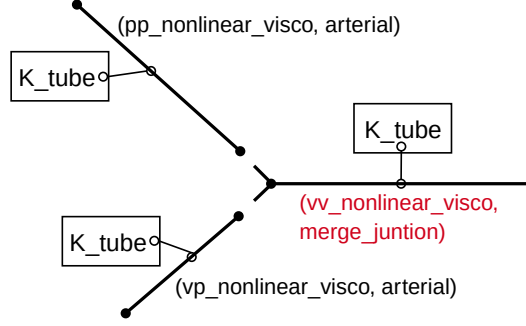
- The parameters that need to be included in the model are given to you in the following file:

```
{Ca_user_dir}/resources/extra_parameters_LeftArm.csv
```

which you can copy and paste into the `parameters.csv` file.



(a) Example of vessel modules to use to define a **bifurcation**.



(b) Example of vessel modules to use to define a **merge junction**.

Figure 3: Schematic of some vessel modules to model junctions. Each vessel module is defined as the pair “(BC_type, vessel_type)”, and it is also coupled to a “K_tube” module to compute its material properties (e.g., the stiffness parameter K in the nonlinear elastic tube law (1)).

- You will need to use a combination of “arterial”, “split_junction” (1-in-2-out) and “merge_junction” (2-in-1-out) to model the superficial palmar arch loop (see Figure 3).
- BC types must be compatible, an output v (e.g., $pv_nonlinear_visco$) must connect to an inlet p (e.g., $pp_nonlinear_visco$ or $pv_nonlinear_visco$) and vice versa. Use $nonlinear_visco$ for all.
- Currently, we enforce that the split/merge side of a vessel needs to have a flow boundary condition, therefore, $merge_junction$ must be vv or vp type and $split_junction$ must be vv or pv type.
- Re-calculate estimates of terminal resistance R_T for $\{ arm_L_T, digital_L_1_T, digital_L_2_T \}$ according to the following formula

$$R_{T_j} = \frac{P_{art} - P_{ven}}{CO * BFF_j}. \quad (2)$$

Assume that the desired cardiac output in the model is $CO = 100 \cdot 10^{-6} \text{ m}^3/\text{s}$, estimated mean arterial and venous pressures are $P_{art} = 12000 \text{ Pa}$ and $P_{ven} = 666 \text{ Pa}$, respectively, and the desired fractional flows to the three terminals are:

Terminal	BFF (%)	C_T ($\text{m}^3 \cdot \text{Pa}^{-1}$)
arm_L_T	1.75%	$1 \cdot 10^{-10}$
digital_L_1_T	0.875%	$1 \cdot 10^{-11}$
digital_L_2_T	0.875%	$1 \cdot 10^{-11}$

Table 1

Set the terminal compliance to be small for now (see Table 1). These parameters could be further calibrated to PPG data.

7. Again, navigate to `{project_dir}/user_run_files/` and do the following command to generate the model with your new arm circulation connected.

```
./run_autogeneration.sh
```

8. Open the generated model and plot the digital_L2 pressure.
9. Extra: Navigate to {project_dir}/user_run_files/ and do the following command to run a parameter identification algorithm on the generated model:

```
./run_param_id.sh {number_of_cores}
```

this will run the optimisation algorithm defined in

```
{CA_user_dir}/simple_physiological_user_inputs.yaml
```

to calibrate the parameters defined in

```
{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_params_for_id.csv
```

to the ground truth values defined in

```
{Ca_user_dir}/resources/simple_physiological_nonlinear_visco_obs_data.json
```

You can run the following command to plot and compare the model outputs with the ground truth observables:

```
./plot_param_id.sh
```

All identified parameter information and plots are stored in

```
{Ca_user_dir}/param_id_output/
```

For more information on using Circulatory Autogen for parameter identification, see https://finbarargus.github.io/circulatory_autogen/parameter-identification/

4 Cell Modelling With Bond Graph Tutorial

1. Navigate to the VITAL-horizoneurope/trainingschool03/smc_with_piezo/ directory. We will now call this dir {CA_user_dir}" and the Circulatory Autogen directory {project_dir}.
 - Reminder, the CA_user_dir contains input files that define the modules that are used in your models and their parameters (and later the information for parameter identification).

Open {project_dir}/user_run_files/user_inputs.yaml and set

```
user_inputs_path_override: {CA_user_dir}/smc_with_piezo_user_inputs.yaml
```

This model is for the smc by Hernandez Hernandez, which we have replicated in CellML.

2. Navigate to {project_dir}/user_run_files/ in terminal and do the following command to generate the model.

```
./run_autogeneration.sh
```

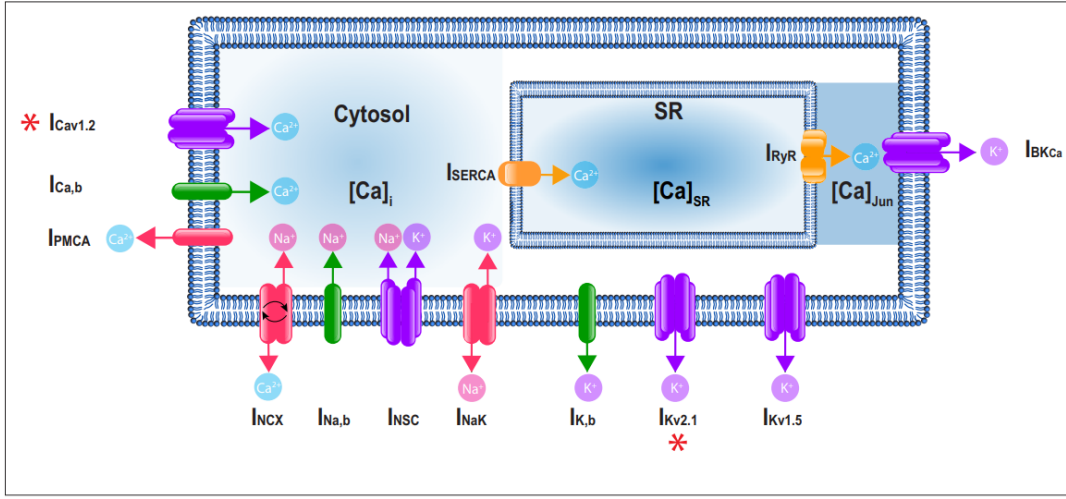


Figure 1. A schematic representation of the Hernandez-Hernandez model. The components of the model include major ion channel currents shown in purple including the voltage-gated L-type calcium current (I_{Ca}), nonselective cation current (I_{NSC}), voltage-gated potassium currents ($I_{Kv1.5}$ and $I_{Kv2.1}$), and the large-conductance Ca^{2+} -sensitive potassium current (I_{BKCa}). Currents from pumps and transporters are shown in red including the sodium/potassium pump current (I_{NaK}), sodium/calcium exchanger current (I_{NCX}), and plasma membrane ATPase current (I_{PMCA}). Leak currents are indicated in green including the sodium leak current ($I_{Na,b}$), potassium leak current ($I_{K,b}$), and calcium leak current ($I_{Ca,b}$). In addition, two currents in the sarcoplasmic reticulum are shown in orange: the sarcoplasmic reticulum Ca-ATPase current (I_{SERCA}) and ryanodine receptor current (I_{RyR}). Calcium compartments comprise three discrete regions including cytosol ($[Ca]_i$), sarcoplasmic reticulum ($[Ca]_{SR}$), and the junctional region ($[Ca]_{Jun}$). Red stars (*) indicate measured sex-specific differences in ionic currents.

Obtained from G. Hernandez-Hernandez et al., “A computational model predicts sex-specific responses to calcium channel blockers in mammalian mesenteric vascular smooth muscle,” eLife, vol. 12, p. RP90604, Feb. 2024, doi: 10.7554/eLife.90604.

The model will be generated in

`{CA_user_dir}/generated_models/smc_with_piezo`

3. Open the generated model in OpenCOR (`smc_with_piezo.cellml`) and plot the membrane potential, V . Quadruple the sodium leak channel conductance ($G_{Na,b}$) to investigate how membrane voltage is affected by increasing leak sodium current.
4. Now we are going to include a piezo ion channel to allow us to model the response to stretch.

The equations for current and gating of the piezo channel we will model are:

$$I_{piezo} = g m_V m_{stretch} (V_m - E_{Na})$$

$$\frac{\partial m_V}{t} = \frac{m_V^\infty - m_V}{\tau_V}$$

$$m_V^\infty = \frac{1}{1 + e^{-(V_m - V_{50})/k_V}}$$

$$\frac{\partial m_{stretch}}{t} = \frac{m_{stretch}^\infty - m_{stretch}}{\tau_{stretch}}$$

$$m_{stretch}^\infty = \frac{1}{1 + e^{-(stretch_m - stretch_{50})/k_{stretch}}}$$

Draw the BG diagram for the piezo channel (stretch can be an external input) and how it connects to the rest of the model, departing from the schematic in Figure 4.

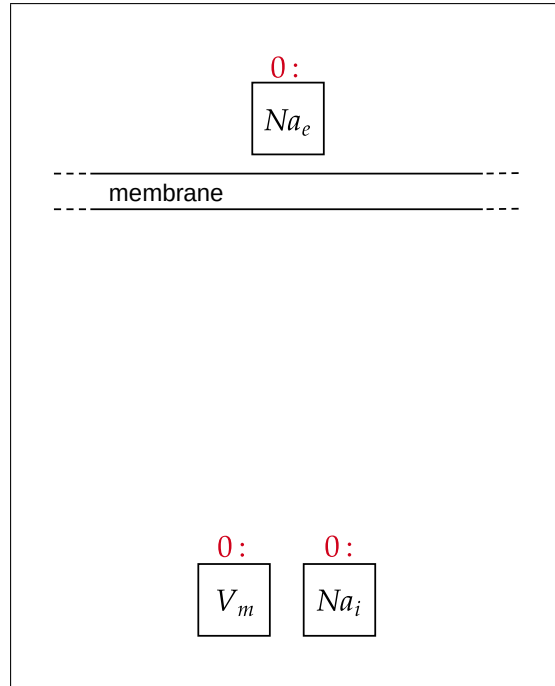


Figure 4: Start of the Piezo BG diagram.

5. Derive the conservation of charge equations for the piezo channel and all of the other currents I_{other} .
 6. Derive the conservation of molar amount equations for the piezo channel and all other Na currents $I_{Na-other}$.
 7. Open the file `{project_dir}/module_config_user/tutorial_modules.cellml` within OpenCOR and edit the module named `i_piezo_Na_tutorial` to include the stretch sensitive equations for the piezo channel:
 8. Open the file `{project_dir}/module_config_user/tutorial_module_config.json` and edit the `I_piezo_Na_tutorial` module to have the stretch sensitive parameters from the cellml model in "variables_and_units" and a general port for I_{piezo} . Copy the naming convention of the corresponding port in the `smc_hernandez_with_piezo`
 9. Open the file `{project_dir}/module_config_user/tutorial_modules.cellml` again in OpenCOR and edit the module named `smc_hernandez_with_piezo` to include the piezo current in the membrane voltage calculation (conservation of charge) and the Na concentration calculation (conservation of molar amount).
 10. Now open `{CA_user_dir}/resources/smc_with_piezo_vessel_array.csv` and add a row for the piezo channel module, with corresponding connections.
- All model parameter values are given to you in `{CA_user_dir}/resources/smc_with_piezo_parameters.csv`
11. Finally, repeat step 3 to generate the model.
 12. Open the generated model in OpenCOR (`Ca_user_dir/generated_models/smc_with_piezo/smc_with_piezo.cellml`) and investigate how changing the stretch ratio r affects the resting membrane potential V_m and the sodium concentration Na_i .

5 Blood Flow - Cellular Coupling Tutorial

Now we will couple the SMC model that we developed in the tutorial from Section 4 with a vessel module within our circulatory system model from Section 3, so that the aortic vessels incorporate myogenic effects.

1. Open the file `{project_dir}/module_config_user/tutorial_modules.cellml` in OpenCOR and navigate to the `material_prop_visco_myogenic_type` module. Here, we will develop a module that calculates a changing stiffness, K_{tube} , dependent on the SMC Ca concentration.

- **IMPORTANT NOTE:** This simple module will not take into account myofiber mechanics, vessel geometry effects (not to mention that Piezo channel effects may be mostly through endothelial cell-SMC gap junction interactions), etc. This tutorial is used to demonstrate how cell models can be coupled to vessel models in this framework. The development of physiologically accurate control of vessel constriction from SMC dynamics will be done within VITAL.

2. Add a calculation of the stretch ratio r_{ratio} in the `material_prop_visco_myogenic_type` module.

$$r_{ratio} = r/r_{us}$$

3. Now include calculation of K_{Ca} as below.

$$K_{Ca_{mod}} = 1 + \frac{1}{1 + e^{-(Ca - Ca_{50})/k_{Ca}}}$$

This is a sigmoid that assumes Ca increases can only increase stiffness up to a maximum of 2x. Ca_{50} sets the midpoint of the sigmoid and k_{Ca} sets the steepness of the sigmoid.

4. Open `{project_dir}/module_config_user/tutorial_module_config.json` and add a general port named `Ca_port` with Ca as the port variable to `material_prop_visco_myogenic` so that `K_tube_calc` and `smc` model can couple together. Note: Look at the ports in `smc_hernandez_with_piezo` within `tutorial_module_config.json` to see that there is a corresponding `Ca_port`. Modules with corresponding ports are coupled with the corresponding port variables.
5. Navigate to the `20250321-day04/coupled_CVS_smc` dir. We will now call this dir `{CA_user_dir}`
6. Now the modules need to be correctly connected in your desired model. Add a `smc` entry and `piezo\channel` entry to the `coupled_CVS_smc_vessel_array.csv` and make sure the `smc` is an output of `ascending_aorta_A` and `smc` has an input of `ascending_aorta_A` and an output of `piezo_channel`. We have supplied you with the parameters required for the coupled model in `coupled_CVS_smc_parameters.csv`.
7. As in previous tutorials, open `{project_dir}/user_run_files/user_inputs.yaml` and set

`user_inputs_path_override: {CA_user_dir}/coupled_CVS_smc_user_inputs.yaml`

8. Navigate to `{project_dir}/user_run_files/` and execute the following command to generate the coupled model:

`./run_autogeneration.sh`

9. Open the generated model (`Ca_user_dir/generated_models/coupled_CVS_smc/coupled_CVS_smc.cellml`) in OpenCOR and investigate how changing the Ca midpoint Ca_{50} and the sigmoid steepness k_{Ca} affects the pressure p and flow v profiles in the `ascending_aorta_A`.